Mapping Yields. Or: How I Learned Kriging is Fun

If you've ever asked a farmer which parts of a paddock perform well, you'll often get an answer like:

"Yeah, the low patch is always wet, the corner under the big tree dries out unless we get July frosts, and the rest is alright."

This info is not particularly useful, unless... we put it in a map (not treasure map unfortunately, but still pretty awesome). I'm talking about yield maps, digital maps that show how much is harvested in every corner of a field. These maps are key tools in precision agriculture, where the goal is to give each bit of land exactly the care it needs.

Sounds straightforward: strap a yield sensor to a combine harvester and boom, instant map. Well… not quite.

The Data Drama

Think of it as a 20,000-piece jigsaw puzzle where some pieces are upside down, some are duplicates, and a few belong to a completely different puzzle. That's what raw yield data looks like. Before any mapping can happen, you have to clean it up:

- Filter out weird sensor spikes,
- Adjust for the combine's stopping and starting,
- Make sure multiple years of data actually line up.

This step is less "GIS magic" and more "serious data scrubbing." But without it, the maps are simply useless. In ArcGIS Pro, the key tools are (see Figure 1):

- Select Layer by Location, to remove points that are outside the field boundary.
- Select Layer by Attribute, to remove points with extremely high or low values or invalid moisture measurements.
- Copy Features, to save Layers as Features.

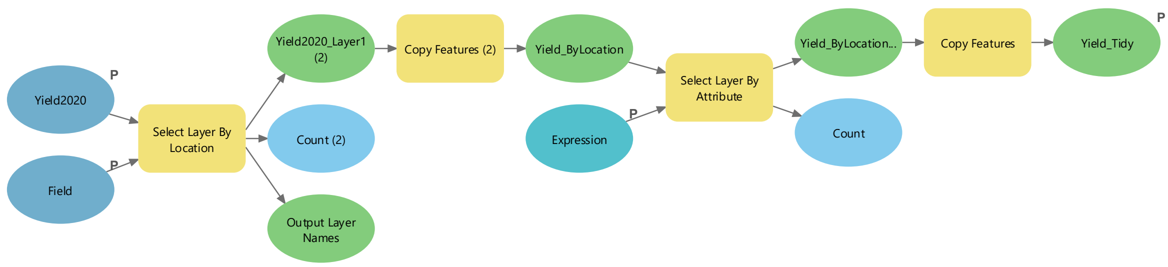

Figure 1: Flow Chart to Tidy Yield Points

And here we go! Figure 2 shows a beautiful and tidy layer of yield points (at least way tidier than the original points). Arguably there are still a few points that seem to be odd, but it definitely does a good job of removing extreme outliers.

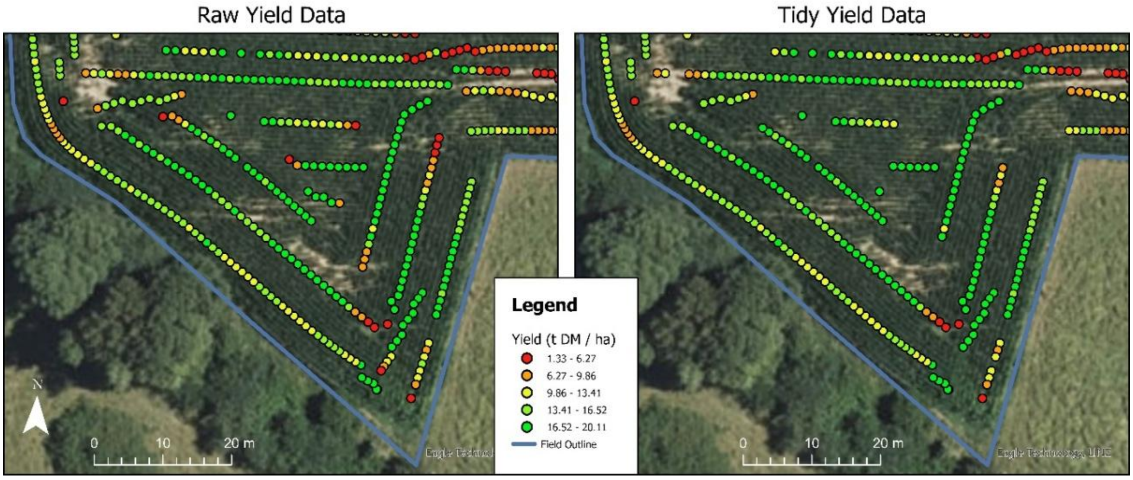

Figure 2: Raw Yield Data and Tidy Yield Data

Interpolation: Filling in the Blanks

Once the messy data is cleaned, the next question is: how do you fill in the gaps between the harvester's GPS tracks? This is where interpolation methods come in.

In ArcGIS, I experimented with several common methods. Deterministic approaches like IDW and Natural Neighbours, polynomial and spline-based techniques, as well as geostatistical methods such as Kriging and Empirical Bayesian Kriging. Using the Geostatistical Analyst toolbox, I applied each interpolation method with default parameters to generate surface maps. As Figure 3 shows, the differences were immediately visible: some methods produced smooth, almost carpet-like surfaces, while others emphasized local variability so strongly that the fields looked like a map of spilled spaghetti sauce.

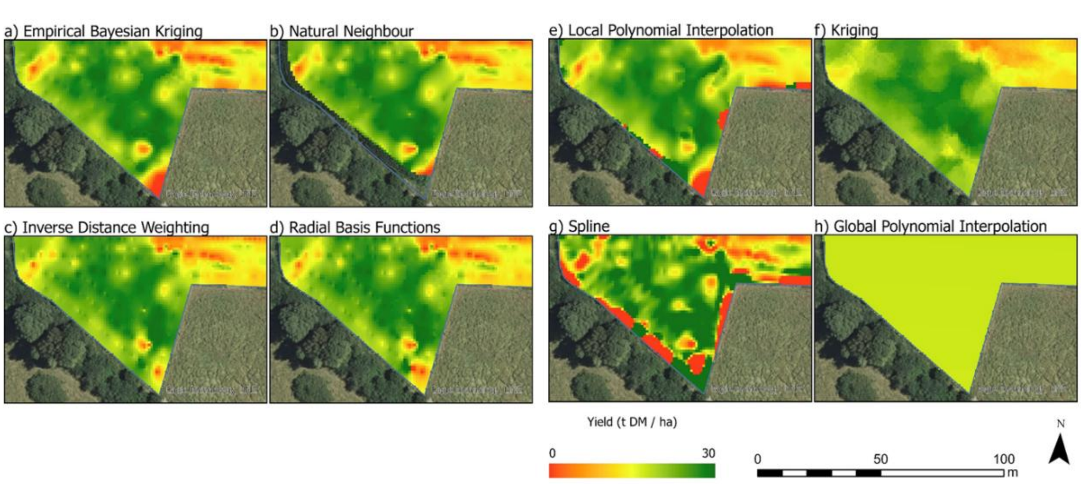

Figure 3: Interpolated Maps

The Big Winner: Kriging

After all the statistical testing, visual checks, and a lot of head-scratching, one method stood out: Kriging. Why?

- Produce surfaces that are smooth but still reflect local variability.
- Explicitly model spatial dependence using a semivariogram.
- Provide error estimates that inform decision-making.
- And (bonus points) it's a fancy word that sounds super professional.

So, if you want reliable yield maps, Kriging is your friend.

Brilliant, but Kinda High-Maintenance

Kriging is powerful, but it isn't a silver bullet. Some important limitations include:

- Semivariogram sensitivity. The entire interpolation depends on the choice of the semivariogram model. A poor fit can produce misleading results.
- Underlying assumptions. Kriging assumes that statistical properties (mean, variance) are consistent across the study area. Yield data often violates this, especially in heterogeneous paddocks.
- Computational intensity. Classic Kriging can be slow for large datasets with tens of thousands of yield points.
- False confidence. The maps look smooth and precise, but the interpolation can oversimplify sharp, real-world changes (e.g., a sudden soil texture shift).

In short: Kriging is excellent when carefully applied, but it requires good data cleaning, careful semivariogram fitting, and a healthy dose of scepticism.

Lessons Learned

- GIS is not just for urban planners and geographers. It's a game-changer in agriculture.
- Clean data is everything. No amount of fancy math can fix bad inputs.
- Sometimes, the method with the weirdest name really is the best.

And next time you see a combine rolling through a Canterbury paddock, don't forget: Behind the scenes, there's a whole lot of geospatial magic turning that noisy machine data into maps that help farmers make smarter decisions.

For Further Reference:

Eagle Technology Group. (2025a). *Comparing interpolation methods*. Retrieved 05/08/2025 from https://pro.arcgis.com/en/pro-app/latest/tool-reference/3d-analyst/comparing-interpolation-methods.htm

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